

Artificial Intelligence for Virtual Characters – IAPV (2025/2026)

Practical project statement – Part 3 (v1.0)

- v1.0 (21/11/2025): Initial Version.

General Information

- This document presents a set of instructions for completing **Part 3** of the Practical Project in the form of independent work by the groups formed for Part 1 of the project, in accordance with the rules indicated in class.
- It should be noted that although the project statement will be made available in parts, the project will be delivered as a whole by the group at a single point in time.
- The Practical Project as a whole must be defended in an oral discussion session, where all members of the group must be present; all members of the group must be able to explain all elements submitted by the group.
- Whenever deemed necessary, tips will be added to Moodle alongside this document. Careful completion and analysis of the various laboratories is essential for the completion of this work.
- Some tips may be provided during theoretical, practical, and laboratory classes, so it is essential that at least one member of the group is always present at all classes.
- Sources available on the internet (e.g., manuals, tutorials, YouTube videos, ChatGPT) may be consulted as a source of learning and inspiration, but the entire project (analysis, design, and implementation) must be authored by the group.
- The entire project, from its conception, implementation, validation, and reporting, must be carried out solely and exclusively by members of the group, i.e., without any recourse to other persons or services.
- In order to maximise the learning process and ensure that the group is heading in the right direction, students are advised to seek feedback from teachers during the development of the project, using classes and question time, **but not by e-mail**.

Objectives

In Part 3 of the practical project, groups will have the opportunity to explore the use of imitation learning algorithms known as Behavioural Cloning (BC) and Generative Adversarial Imitation Learning (GAIL). Part 3 of the project will contribute to approximately **40% of the final mark** awarded for the Project. This percentage is only indicative, serving to enable the group to plan their development efforts, and may be changed when the remainder of the project becomes available.

The code development will be based on the Python imitation package. On the website for this package, <https://imitation.readthedocs.io/en/latest/index.html>, you can find a detailed description of its API, as well as multiple examples of ready-to-run code. Of particular interest to this project:

- https://imitation.readthedocs.io/en/latest/tutorials/1_train_bc.html
- https://imitation.readthedocs.io/en/latest/tutorials/3_train_gail.html
- https://imitation.readthedocs.io/en/latest/tutorials/10_train_custom_env.html
- <https://imitation.readthedocs.io/en/latest/main-concepts/trajectories.html>
- <https://imitation.readthedocs.io/en/latest/main-concepts/experts.html>

Based on the information and code collected from the imitation package page, groups will need to complete the following steps to finish Part 3 of the practical project:

1. Create a set of Python files that can be invoked from the terminal, receiving a set of input arguments:
 - a. Create a Python programme **train.py** that:
 - i. Receives the path to a file with a set of demonstrations D;
 - ii. Receives the path to an output file O where the training result will be saved;
 - iii. Receives the name of the gym G where D was collected: CartPole (consider “seals/CartPole-v0”) or Custom;
 - iv. Receives the definition of an imitation learning algorithm A: BC or GAIL;
 - v. Trains a policy P to approximate D, using A in G;
 - vi. Saves P to file O.
 - vii. Tips:
 1. Example of a call in the terminal:

```
python3 train.py --file="demo_1.pkl" --gym="CartPole" --algorithm="GAIL" --output="policy.zip"
```
 - b. Create a Python programme **run.py** that:
 - i. Receives the path to a policy file P;
 - ii. Receives the name of the gym G where P was trained: CartPole or Custom;
 - iii. Opens G in visualisation mode;
 - iv. Runs P in G step by step or in continuous mode, according to the user's choice and for as long as they wish.
 - c. Create a Python programme **demos.py** that:
 - i. Receives the name of a gym G: CartPole or Custom;

- ii. Receives a number of demonstration episodes E ;
 - iii. Receives the path to a file F where it will save the demonstrations;
 - iv. Receives a flag to choose whether to use a pre-trained “ppo-huggingface” policy to generate the set of demonstrations D (only for the CartPole case);
 - v. Opens G in visualisation mode;
 - vi. Obtains demonstrations through pre-trained policy (if option enabled) or by asking the user to control the agent in G with the keyboard to perform the task during E ;
 - vii. Stores the state-action pairs produced during E in D ;
 - viii. Saves D in F .
2. Create a new **Custom** gym that can be used in `train.py`, `run.py`, and `demos.py` with the following characteristics:
- a. The environment is a regular $n \times m$ grid (gym parameters);
 - b. There is a randomly chosen target cell;
 - c. There is a set of k cells, chosen at random, considered walls;
 - d. The agent's actions are to move one square sideways or vertically;
 - e. The environment cannot allow the agent to pass through walls;
 - f. The agent observes its own position (i.e., the cell where it is located), the presence/absence of walls in the cells adjacent to its position, and the relative position of the objective in relation to the agent's position;
 - g. The environment is visualised in text mode on the terminal;
 - h. The agent's task is to reach the objective O as quickly as possible.
3. Conduct a series of experiments:
- a. Create files with demonstrations using `demos.py` for the CartPole and Custom environments, requesting actions from the user and, in the case of CartPole, also through pre-trained policy;
 - b. Train the agents in both environments, using the stored demonstrations and resorting to the BC and GAIL algorithms;
 - c. Compare the performance of the two learning algorithms in both test environments, taking into account multiple training and execution runs.
4. Produce a report that:
- a. Describes in detail the code developed and the experiments carried out;
 - b. Presents the data obtained in the experiments in the form of tables and graphs;
 - c. Includes a critical analysis of the results obtained;
 - d. Includes links to videos produced by the group, uploaded to YouTube as *unlisted*, where the trained agents can be seen in action.

Mandatory deliverables for project to be accepted for evaluation

- Final delivery
 - Regarding part 3:
 - Link to .zip file containing:
 - .word and .pdf files of the report (with links to videos included).
 - Code developed by the group.
 - Demonstration files used to train the agents.
 - The document containing the group's self-assessment (*template* provided in Moodle), which will include a description of each group member's involvement (what and how much each one did); the grade will be differentiated between group members.

Assessment criteria

- Degree of compliance with the objectives and steps stipulated for the project, including the various requirements and quality criteria presented in the description of each of the steps.
- Level of student autonomy.
- Quality of performance (group and individual) in the oral discussion of the project.
- Although the Practical Project is developed in groups, the mark given to each student in the group is individualised according to the student's contribution to the Practical Project and the student's performance in the oral discussion.

Final delivery deadline

- Mandatory deliverables must be sent by email to pedro.santana@iscte-iul.pt and sancho.oliveira@iscte-iul.pt approximately one week before the date set for the oral discussions (exact date to be determined in due course).
- The oral discussions will take place on the date assigned for the first assessment period, to be determined by Iscte; as soon as Iscte determines the date for the first assessment period, a new version of this document will be produced with the relevant information.

Plagiarism

- All elements to be included in the project, including narrative, storyboard, characters, modelling, and animation, must be original, unique, invented, and developed by the group. Failure to comply with this requirement will be considered plagiarism.
- The project is to be developed entirely by the group members, without access to or sharing of project-specific information with individuals outside the group, with the exception of the lecturer.
- Detection of plagiarism will result in disciplinary proceedings, as recommended by the ISCTE-IUL Pedagogical Council.